

2015 HSC Mathematics Marking Guidelines

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Section I

Multiple-choice Answer Key

Question	Answer
1	D
2	C
3	A
4	A
5	B
6	C
7	B
8	C
9	D
10	A

Section II

Question 11 (a)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}
 &4x - (8 - 6x) \\
 &= 4x - 8 + 6x \\
 &= 10x - 8
 \end{aligned}$$

Question 11 (b)

Criteria	Marks
• Provides correct solution	2
• Finds common factor of 3, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 &3x^2 - 27 \\
 &= 3(x^2 - 9) \\
 &= 3(x - 3)(x + 3)
 \end{aligned}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Attempts to use $2 - \sqrt{7}$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \frac{8}{2 + \sqrt{7}} &= \frac{8}{2 + \sqrt{7}} \times \frac{2 - \sqrt{7}}{2 - \sqrt{7}} \\
 &= \frac{8(2 - \sqrt{7})}{4 - 7} \\
 &= \frac{8(2 - \sqrt{7})}{-3} \\
 &= \frac{8\sqrt{7} - 16}{3}
 \end{aligned}$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Identifies the common ratio, or equivalent merit	1

Sample answer:

$$a = 1 \quad r = -\frac{1}{4}$$

$$\begin{aligned}
 S &= \frac{a}{1-r} \\
 &= \frac{1}{1 - \left(-\frac{1}{4}\right)} \\
 &= \frac{4}{5}
 \end{aligned}$$

Question 11 (e)

Criteria	Marks
• Provides correct derivative	2
• Attempts to use chain rule, or equivalent merit	1

Sample answer:

$$\text{Let } y = (e^x + x)^5$$

$$\frac{dy}{dx} = 5(e^x + x)^4 \times (e^x + 1)$$

$$= 5(e^x + 1)(e^x + x)^4$$

Question 11 (f)

Criteria	Marks
• Provides correct derivative	2
• Attempts to use product rule, or equivalent merit	1

Sample answer:

$$\text{Let } u = x + 4 \quad v = \ln x$$

$$u' = 1 \quad v' = \frac{1}{x}$$

$$\therefore y' = u'v + v'u$$

$$= 1 \cdot \ln x + \frac{1}{x}(x + 4)$$

$$= \ln x + \frac{x + 4}{x}$$

Question 11 (g)

Criteria	Marks
• Provides correct solution	2
• Provides correct primitive, or equivalent merit	1

Sample answer:

$$\int_0^{\frac{\pi}{4}} \cos 2x \, dx = \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \sin \frac{\pi}{2} - \frac{1}{2} \sin 0$$

$$= \frac{1}{2} - 0$$

$$= \frac{1}{2}$$

Question 11 (h)

Criteria	Marks
• Provides correct primitive	2
• Recognises $\frac{f'(x)}{f(x)}$, or equivalent merit	1

Sample answer:

$$\begin{aligned}\int \frac{x}{x^2 - 3} dx \\&= \frac{1}{2} \int \frac{2x}{x^2 - 3} dx \\&= \frac{1}{2} \ln(x^2 - 3) + C\end{aligned}$$

Question 12 (a)

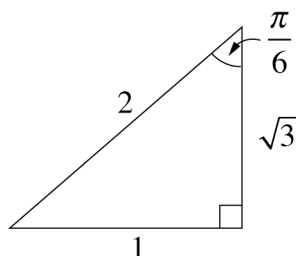
Criteria	Marks
• Provides correct solutions	2
• Provides one correct solution, or equivalent merit	1

Sample answer:

$$2 \sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

**Question 12 (b) (i)**

Criteria	Marks
• Provides correct solution	2
• Attempts to use the fact that the diagonals are perpendicular, or equivalent merit	1

Sample answer:

$$\ell_1 \text{ is } y = m_1 x + b$$

$$\ell_2 \text{ is } y = -\frac{x}{3} \Rightarrow m_2 = -\frac{1}{3}$$

$$\ell_1 \perp \ell_2 \quad OABC \text{ rhombus (diagonals } \ell_1, \ell_2 \text{ are perpendicular)}$$

$$m_1 m_2 = -1$$

$$m_1 \times \left(-\frac{1}{3}\right) = -1$$

$$\therefore m_1 = 3$$

$$\ell_1 \text{ is } y = 3x + b$$

$$11 = 3 \times 7 + b$$

$$b = -10$$

$$\therefore y = 3x - 10$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct coordinates	2
• Makes some progress towards correct answer	1

Sample answer:

Solving $y = -\frac{x}{3}$

and $y = 3x - 10$ simultaneously

$$-\frac{x}{3} = 3x - 10$$

$$x = -9x + 30$$

$$10x = 30$$

$$x = 3$$

Hence, $y = -\frac{3}{3}$

$$= -1$$

$$\therefore D(3, -1)$$

Question 12 (c)

Criteria	Marks
• Provides correct derivative	2
• Attempts to use the quotient rule, or equivalent merit	1

Sample answer:

Let $u = x^2 + 3$ $v = x - 1$

$$u' = 2x \quad v' = 1$$

$$y' = \frac{u'v - v'u}{v^2}$$

$$= \frac{2x(x-1) - 1(x^2 + 3)}{(x-1)^2}$$

$$= \frac{2x^2 - 2x - x^2 - 3}{(x-1)^2}$$

$$= \frac{x^2 - 2x - 3}{(x-1)^2}$$

Question 12 (d)

Criteria	Marks
• Provides correct solution	2
• Evaluates discriminant, or equivalent merit	1

Sample answer:

For real roots, $\Delta \geq 0$

$$\Delta = b^2 - 4ac$$

$$= (-8)^2 - 4(1)(k)$$

$$= 64 - 4k$$

$$\therefore 64 - 4k \geq 0$$

$$64 \geq 4k$$

$$k \leq 16$$

Question 12 (e) (i)

Criteria	Marks
• Provides a correct equation	2
• Finds correct gradient, or equivalent merit	1

Sample answer:

$$y = \frac{x^2}{2} \quad P\left(1, \frac{1}{2}\right)$$

$$\frac{dy}{dx} = x$$

$$\text{when } x = 1, \quad m = 1$$

The equation of the tangent is

$$y - y_1 = m(x - x_1)$$

$$y - \frac{1}{2} = 1(x - 1)$$

$$\therefore y = x - \frac{1}{2}$$

Question 12 (e) (ii)

Criteria	Marks
• Provides a correct equation	1

Sample answer:

$$\text{Focus } S(0, a) = \left(0, \frac{1}{2}\right) \therefore a = \frac{1}{2}$$

$$\text{Equation directrix is: } y = -a$$

$$\therefore y = -\frac{1}{2}$$

Question 12 (e) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\text{Tangent: } y = x - \frac{1}{2}$$

$$\text{Directrix: } y = -\frac{1}{2}$$

$$\begin{aligned} \text{Solution: } -\frac{1}{2} &= x - \frac{1}{2} \\ x &= 0 \end{aligned}$$

$\therefore Q$ lies on y-axis

Question 12 (e) (iv)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$SQ = SO + OQ$$

$$= \frac{1}{2} + \frac{1}{2}$$

$$= 1$$

$$PS = 1$$

$$\therefore SQ = PS$$

$\therefore \triangle PQS$ is isosceles (two equal sides)

Question 13 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos A = \frac{8^2 + 6^2 - 4^2}{2 \times 8 \times 6}$$

$$= \frac{84}{96}$$

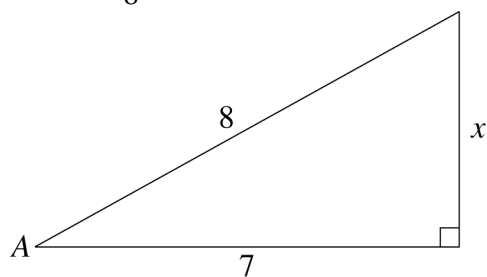
$$= \frac{7}{8}$$

Question 13 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds exact value of $\sin A$, or equivalent merit	1

Sample answer:

$$\cos A = \frac{7}{8}$$



$$x^2 = \sqrt{8^2 - 7^2}$$

$$x = \sqrt{15}$$

$$\therefore \sin A = \frac{\sqrt{15}}{8}$$

$$\therefore \text{Area } \triangle ABC = \frac{1}{2}bc \sin A$$

$$= \frac{1}{2} \times 8 \times 6 \times \frac{\sqrt{15}}{8}$$

$$\therefore \text{Area} = 3\sqrt{15} \text{ cm}^2$$

Question 13 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Identifies -3 and 3 as important values, or equivalent merit	1

Sample answer:

$$\text{Domain: } 9 - x^2 \geq 0$$

$$x^2 \leq 9$$

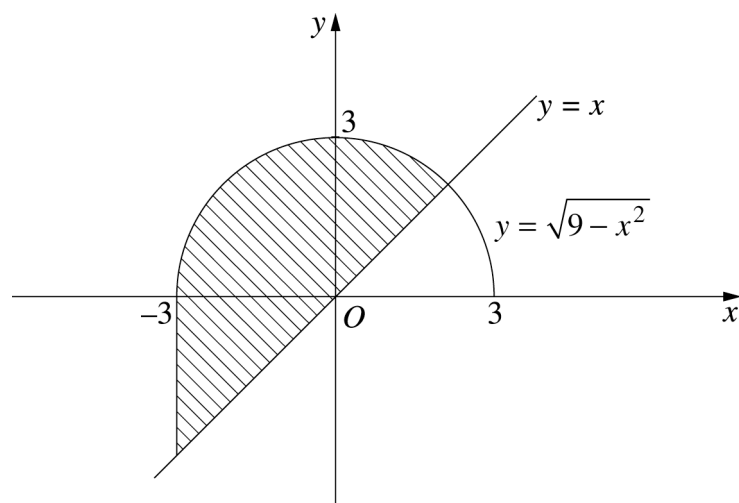
$$-3 \leq x \leq 3$$

$$\text{Range: } 0 \leq y \leq 3$$

Question 13 (b) (ii)

Criteria	Marks
• Shades correct region	2
• Shades a region involving the correct semicircle, or equivalent merit	1

Sample answer:



Question 13 (c) (i)

Criteria	Marks
• Provides correct solution	4
• Finds coordinates of both stationary points, or equivalent merit	3
• Factorises quadratic in $\frac{dy}{dx} = 0$ correctly, or equivalent merit	2
• Attempts to solve $\frac{dy}{dx} = 0$, or equivalent merit	1

Sample answer:

$$y = x^3 - x^2 - x + 3$$

$$y' = 3x^2 - 2x - 1$$

$$y'' = 6x - 2$$

Stationary points when $y' = 0$ ie $3x^2 - 2x - 1 = 0$

$$(3x + 1)(x - 1) = 0$$

$$\therefore x = -\frac{1}{3} \text{ or } x = 1$$

when $x = -\frac{1}{3}$, $y = 3\frac{5}{27}$ and $y'' = -4 < 0$ (cd)

$\therefore \left(-\frac{1}{3}, 3\frac{5}{27}\right)$ is a local maximum

when $x = 1$, $y = 2$ and $y'' = 4 > 0$ (cu)

$\therefore (1, 2)$ is a local minimum

Question 13 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Shows second derivative = 0 at $x = \frac{1}{3}$, or equivalent merit	1

Sample answer:

$$P\left(\frac{1}{3}, \frac{70}{27}\right)$$

Point of inflexion where $y'' = 0$ and concavity changes

ie $6x - 2 = 0$

$$x = \frac{1}{3}$$

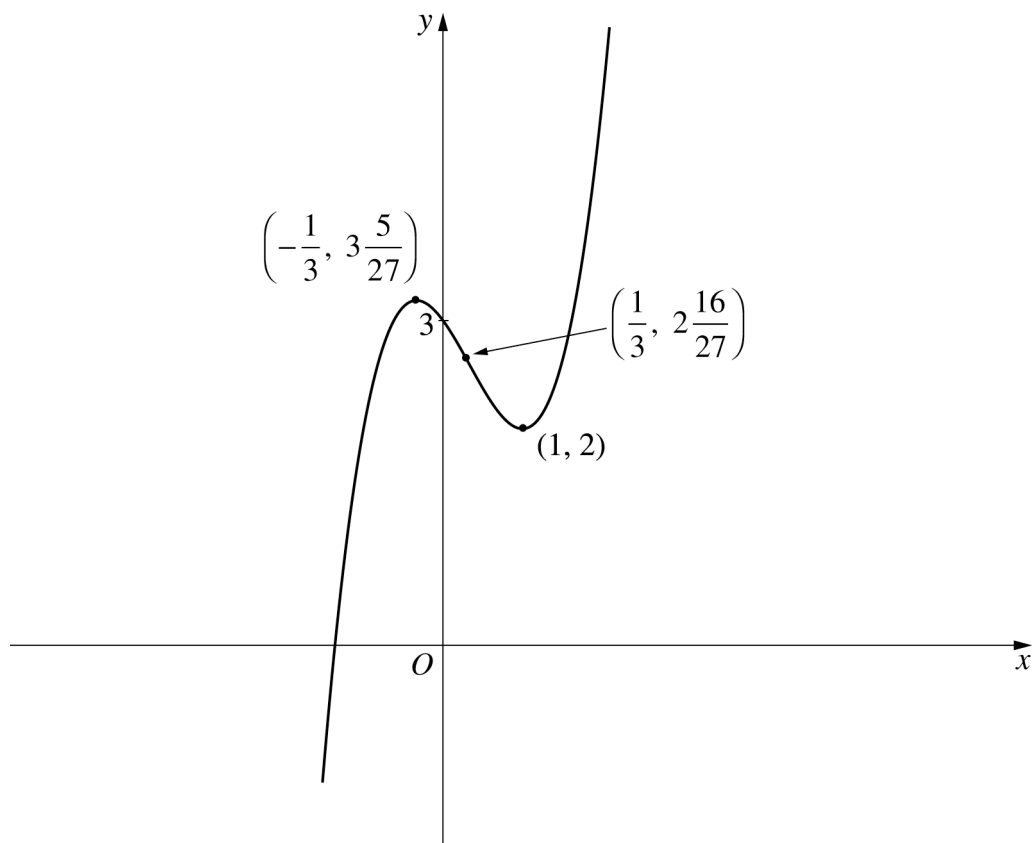
There is a point of inflexion at $x = \frac{1}{3}$ since from (i) concavity changes (maximum at

$x = -\frac{1}{3}$ and minimum at $x = 1$).

Question 13 (c) (iii)

Criteria	Marks
• Provides correct sketch	2
• Correctly plots and labels stationary points and point of inflexion, or equivalent merit	1

Sample answer:



Question 14 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Provides correct expression for $\dot{x}$ including determination of the constant, or equivalent merit	1

Sample answer:

$$\ddot{x} = -10, \quad t = 0, \quad x = 110, \quad \dot{x} = 0$$

$$\dot{x} = \int -10 \, dt$$

$$= -10t + C_1$$

When $t = 0$, $\dot{x} = 0$

$$\therefore 0 = -10(0) + C_1$$

$$\therefore C_1 = 0$$

$$\therefore \dot{x} = -10t$$

$$\text{Now, } x = \int -10t \, dt$$

$$= -5t^2 + C_2$$

When $t = 0$, $x = 110$

$$\therefore 110 = -5(0)^2 + C_2$$

$$\therefore C_2 = 110$$

$$\therefore x = -5t^2 + 110$$

Question 14 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds correct time when brakes are applied, or equivalent merit	1

Sample answer:When $\dot{x} = -37$

$$-37 = -10t$$

$$t = \frac{37}{10}$$

$$\begin{aligned}\therefore x &= -5\left(\frac{37}{10}\right)^2 + 110 \\ &= 41.55\end{aligned}$$

$$\begin{aligned}\therefore \text{It has fallen } (110 - 41.55) \text{ m} \\ = 68.45 \text{ m}\end{aligned}$$

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$P(\text{Saturday dry}) = P(WD) + P(DD)$$

$$\begin{aligned}&= \frac{1}{2} \times \frac{5}{6} + \frac{1}{2} \times \frac{1}{2} \\ &= \frac{2}{3}\end{aligned}$$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Provides correct expression for probability of one of Saturday or Sunday being wet, or equivalent merit	1

Sample answer: $P(\text{both Saturday and Sunday wet})$

$$= P(WWW) + P(DWW)$$

$$= \frac{1}{2} \times \frac{1}{6} \times \frac{1}{6} + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{6}$$

$$= \frac{1}{72} + \frac{1}{24}$$

$$= \frac{1}{18}$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer: $P(\text{at least one of Saturday and Sunday dry})$

$$= 1 - P(\text{both are wet})$$

$$= 1 - \frac{1}{18}$$

$$= \frac{17}{18}$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$r = 0.6\% \text{ per month}$$

$$= 0.006 \text{ per month}$$

$$A_1 = 100\,000(1.006) - M$$

$$A_2 = A_1(1.006) - M$$

$$= (100\,000(1.006) - M)(1.006) - M$$

$$A_2 = 100\,000(1.006)^2 - M(1.006) - M$$

$$= 100\,000(1.006)^2 - M(1 + 1.006)$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Recognises the pattern in the expression for A_n as a sum, or equivalent merit	1

Sample answer:

$$A_3 = A_2(1.006) - M$$

$$= 100\,000(1.006)^3 - M(1.006)^2 - M(1.006) - M$$

$$\vdots$$

$$A_n = 100\,000(1.006)^n - M[1 + (1.006) + (1.006)^2 + \dots + (1.006)^{n-1}]$$

Now $[1 + (1.006) + (1.006)^2 + \dots + (1.006)^{n-1}]$ isa geometric series, with n terms where $a = 1$ and $r = 1.006$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{1((1.006)^n - 1)}{(1.006) - 1}$$

$$\therefore A_n = 100\,000(1.006)^n - M \left[\frac{1((1.006)^n - 1)}{0.006} \right]$$

Question 14 (c) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$M = 780, n = 120$$

$$A_{120} = 100\,000(1.006)^{120} - 780 \left[\frac{(1.006)^{120} - 1}{0.006} \right]$$

$$= 68\,499.46$$

$\therefore$ amount owing = \$68 500 (nearest \$100)

Question 14 (c) (iv)

Criteria	Marks
• Provides correct solution	3
• Finds correct expression for $(1.006)^n$, or equivalent merit	2
• Attempts to use $A_n = 0$, or equivalent merit	1

Sample answer:

Since the amount owing is \$48 500,

$$A_n = 48\,500(1.006)^n - 780 \left[\frac{(1.006)^n - 1}{0.006} \right]$$

Let $A_n = 0$ when the amount owing is completely repaid

$$\therefore 48\,500(1.006)^n = 780 \left[\frac{(1.006)^n - 1}{0.006} \right]$$

$$\therefore 291(1.006)^n = 780(1.006)^n - 780$$

$$\therefore 489(1.006)^n = 780$$

$$\therefore (1.006)^n = 1.59509\dots$$

$$n \log(1.006) = \log(1.59509\dots)$$

$$n \approx \frac{\log(1.59509)}{\log 1.006}$$

time required ≈ 78.055 months

Question 15 (a) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$C = Ae^{-0.14t}$$

$$\begin{aligned}\frac{dC}{dt} &= -0.14Ae^{-0.14t} \\ &= -0.14C\end{aligned}$$

Question 15 (a) (ii)

Criteria	Marks
• Provides correct value of A	1

Sample answer:When $t = 0$, $C = 130$

$$130 = Ae^0$$

$$\therefore A = 130$$

Question 15 (a) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$C = 130e^{-0.14t}$$

When $t = 7$,

$$C = 130e^{-0.14 \times 7}$$

$$\approx 48.79044285$$

amount of caffeine ≈ 49 mg

Question 15 (a) (iv)

Criteria	Marks
• Provides correct solution	2
• Attempts to use logarithms to solve $65 = 130e^{-0.14t}$, or equivalent merit	1

Sample answer:

Taking $C = 65$

$$65 = 130e^{-0.14t}$$

$$\frac{1}{2} = e^{-0.14t}$$

$$\ln \frac{1}{2} = -0.14t$$

$$t = \frac{\ln \frac{1}{2}}{-0.14}$$
$$= 4.95105129$$

time taken ≈ 4.95 hours

Question 15 (b) (i)

Criteria	Marks
• Provides correct proof	2
• Shows $\angle FDC$ equal to $\angle ADE$, or $\angle ADE$ equal to $\angle DAE$, or equivalent merit	1

Sample answer:

$$\angle FDC = \angle ADE \quad (\text{vertically opposite angles equal})$$

$$\angle ADE = \angle DAE \quad (\text{base angles, isosceles triangle})$$

$$\therefore \angle FDC = \angle DAE$$

$$\begin{aligned} \angle FCD &= \angle ACB \\ &= 90^\circ \quad (\text{given}) \end{aligned}$$

$$\therefore \triangle ACB \parallel \triangle DCF \quad (\text{equal angles})$$

Question 15 (b) (ii)

Criteria	Marks
• Provides correct explanation	1

Sample answer:

$$\angle DFC = \angle ABC \quad (\text{corresponding angles in similar triangles})$$

$$\therefore \triangle EFB \text{ is isosceles} \quad (\text{base angles equal})$$

Question 15 (b) (iii)

Criteria	Marks
• Provides correct solution	2
• Provides correct relationship between lengths of AB and FD, or drops perpendicular from E onto AD, or equivalent merit	1

Sample answer:

$$AB = 2FD \quad (\text{using (i), corresponding sides in proportion } AC = 2DC)$$

$$\text{ie } AE + EB = 2FD$$

$$\text{Also, } FD = EF - ED$$

$$\text{so } FD = EB - AE \quad (\text{by (ii) since } EF = EB)$$

$$\text{Hence } AE + EB = 2(EB - AE)$$

$$\therefore EB = 3AE$$

Question 15 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to solve $\frac{dV}{dt} = 0$, or equivalent merit	1

Sample answer:

$$\text{Set } \frac{dV}{dt} = 0$$

$$80\sin(0.5t) = 0$$

$$\sin 0.5t = 0$$

$$0.5t = 0, \pi, 2\pi, \dots$$

first starts to decrease when $0.5t = \pi$

$$\text{so } t = 2\pi$$

Question 15 (c) (ii)

Criteria	Marks
• Provides correct solution	2
• Provides correct primitive, or equivalent merit	1

Sample answer:

$$\frac{dV}{dt} = 80\sin(0.5t)$$

Integrating,

$$V = -160\cos(0.5t) + C$$

$$\text{When } t = 0, \quad 1200 = -160\cos(0.5 \times 0) + C$$

$$C = 1360$$

$$\text{Hence, } V = -160\cos(0.5t) + 1360$$

$$\text{When } t = 3, \quad V = -160\cos(0.5 \times 3) + 1360$$

$$\text{Volume} \approx 1349 \text{ L}$$

Question 15 (c) (iii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

Greatest volume when $\frac{dV}{dt} = 0$

ie when $t = 2\pi$ and volume starts to decrease

$$\begin{aligned} V &= -160\cos(0.5 \times 2\pi) + 1360 \\ &= 1520 \end{aligned}$$

volume = 1520 L

Question 16 (a) (i)

Criteria	Marks
• Provides correct x -coordinates	1

Sample answer:

$$\text{When } y = 0, x^2 - 7x + 10 = 0$$

$$\Rightarrow (x - 2)(x - 5) = 0$$

$$\therefore x = 2, x = 5$$

So A is $(2,0)$ and B is $(5,0)$

Question 16 (a) (ii)

Criteria	Marks
• Provides correct coordinates	1

Sample answer:

$$\text{When } x = 0, y = 10$$

$$\Rightarrow C \text{ is } (7,10) \quad (\text{by symmetry})$$

Question 16 (a) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\int_0^2 (x^2 - 7x + 10) dx = \left[\frac{x^3}{3} - \frac{7x^2}{2} + 10x \right]_0^2$$

$$= \left(\frac{2^3}{3} - 7 \times \frac{2^2}{2} + 10 \times 2 \right) - \left(\frac{0^3}{3} - 7 \times \frac{0^2}{2} + 10 \times 0 \right)$$

$$= \frac{26}{3}$$

Question 16 (a) (iv)

Criteria	Marks
• Provides correct solution	2
• Attempts to find the difference of two areas or integrals, or equivalent merit	1

Sample answer:

By symmetry, $\int_0^2 (x^2 - 7x + 10) dx = \int_5^7 (x^2 - 7x + 10) dx$

$\therefore$ Shaded area = area of $\triangle - \int_5^7 (x^2 - 7x + 10) dx$

$$= \frac{1}{2} \times (7 - 2) \times 10 - \frac{26}{3}$$

$$= \frac{49}{3} \text{ units}^2$$

Question 16 (b)

Criteria	Marks
• Provides correct solution	3
• Obtains $\pi \int_0^6 \left(e^{\frac{y}{4}} + 2e^{\frac{y}{8}} + 1 \right) dy$, or equivalent merit	2
• Writes down formula for the volume including limits OR • Writes x as a function of y , or equivalent merit	1

Sample answer:

$$V = \pi \int_0^6 x^2 dy$$

$$\text{Now } y = 8 \log_e (x - 1)$$

$$\Rightarrow e^{\frac{y}{8}} = x - 1$$

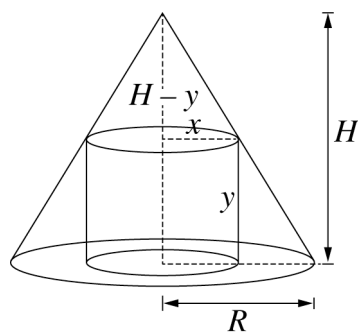
$$\text{ie } x = e^{\frac{y}{8}} + 1$$

$$\begin{aligned} \therefore V &= \pi \int_0^6 \left(e^{\frac{y}{8}} + 1 \right)^2 dy \\ &= \pi \int_0^6 \left(e^{\frac{y}{4}} + 2e^{\frac{y}{8}} + 1 \right) dy \\ &= \pi \left[4e^{\frac{y}{4}} + 16e^{\frac{y}{8}} + y \right]_0^6 \\ &= \pi \left[\left(4e^{\frac{3}{2}} + 16e^{\frac{3}{4}} + 6 \right) - (4 + 16 + 0) \right] \\ &= \pi \left[4e^{\frac{3}{2}} + 16e^{\frac{3}{4}} - 14 \right] \end{aligned}$$

$$\text{volume} \approx 118.7$$

Question 16 (c) (i)

Criteria	Marks
• Provides correct solution	3
• Uses similar triangles to obtain correct equation relating x and y , or equivalent merit	2
• Attempts to use similar triangles, or equivalent merit	1

Sample answer:

Using similar triangles

$$\frac{H-y}{H} = \frac{x}{R} \quad \left[\text{OR,} \quad \frac{H}{R} = \frac{y}{R-x} \right]$$

$$RH - Ry = xH$$

$$RH - xH = Ry$$

$$y = \frac{H(R-x)}{R}$$

$$\therefore V = \pi r^2 h = \pi x^2 y$$

$$= \pi x^2 \frac{H}{R} (R-x)$$

$$= \frac{H}{R} \pi x^2 (R-x)$$

Question 16 (c) (ii)

Criteria	Marks
• Provides correct solution	4
• Finds the value at which the stationary point occurs and verifies it is a maximum, or equivalent merit	3
• Finds the value of x at which $\frac{dV}{dx} = 0$, or equivalent merit	2
• Finds correct expression for $\frac{dV}{dx}$, or equivalent merit	1

Sample answer:

Maximum V occurs when $\frac{dV}{dx} = 0$ and $\frac{d^2V}{dx^2} < 0$

$$V = \frac{H}{R}\pi x^2(R-x) = H\pi x^2 - \frac{H}{R}\pi x^3$$

$$\frac{dV}{dx} = 2H\pi x - \frac{3H}{R}\pi x^2 = 0 \text{ for maximum}$$

$$\therefore 0 = \pi H x \left(2 - \frac{3}{R}x \right)$$

$$x = 0 \text{ (discount)} \text{ or } x = \frac{2R}{3}$$

$$\frac{d^2V}{dx^2} = 2H\pi - \frac{6H}{R}\pi x$$

$$= 2H\pi - \frac{6H}{R}\pi \frac{2R}{3} \text{ at } x = \frac{2R}{3}$$

$$= 2H\pi - 4H\pi$$

$$= -2H\pi < 0 \text{ since } H > 0$$

$$\therefore \text{Maximum volume when } x = \frac{2R}{3}$$

$$\therefore V = \frac{H}{R}\pi \left(\frac{2R}{3} \right)^2 \left(R - \frac{2R}{3} \right)$$

$$= \frac{H}{R}\pi \frac{4R^2}{9} \left(\frac{R}{3} \right)$$

$$V = \frac{4H\pi R^2}{27}$$

2015 HSC Mathematics

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	1.1	P3, P4
2	1	6.2	P4
3	1	7.1	H5
4	1	3.3	H5
5	1	11.3	H4
6	1	13.4, 13.5, 13.7	H5
7	1	11.4	H8
8	1	9.1, 12.1	H3
9	1	14.3	H4, H5
10	1	12.5	H3, H8

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	1	1.3	P3
11 (b)	2	1.3	P4
11 (c)	2	1.1	P3
11 (d)	2	7.3	H5
11 (e)	2	8.9, 12.5	P7, H3
11 (f)	2	8.9, 12.5	P7, H3
11 (g)	2	11.1, 13.7	H5
11(h)	2	11.2, 12.5	H5
12 (a)	2	5.3, 13.1	P4, H5
12 (b) (i)	2	2.3, 6.8	H5
12 (b) (ii)	2	1.4, 6.3, 6.8	P4, H5
12 (c)	2	8.8	P7
12 (d)	2	1.4, 9.2, 9.3	P4
12 (e) (i)	2	6.2, 8.5	P4
12 (e) (ii)	1	9.5	P4, P5
12 (e) (iii)	1	1.4, 6.3	P4, P5
12 (e) (iv)	1	2.2, 6.8	P4, H2, H5
13 (a) (i)	1	5.5	P3, P4
13 (a) (ii)	2	5.5	P3, P4

Question	Marks	Content	Syllabus outcomes
13 (b) (i)	2	4.1	P5
13 (b) (ii)	2	4.1, 4.4	P5
13 (c) (i)	4	10.2, 10.4	H6
13 (c) (ii)	2	10.4	H6
13 (c) (iii)	2	10.5	H6
14 (a) (i)	2	14.3	H4, H5
14 (a) (ii)	2	14.3	H4, H5
14 (b) (i)	1	3.2, 3.3	H5
14 (b) (ii)	2	3.2, 3.3	H5
14 (b) (iii)	1	3.2, 3.3	H5
14 (c) (i)	1	7.5	H4, H5
14 (c) (ii)	2	7.2, 7.5	H4, H5
14 (c) (iii)	1	7.5	H4, H5
14 (c) (iv)	3	7.5, 12.2	H3, H5
15 (a) (i)	1	14.2	H3, H5
15 (a) (ii)	1	14.2	H3
15 (a) (iii)	1	14.2	H3
15 (a) (iv)	2	12.2, 14.2	H3
15 (b) (i)	2	2.3, 2.5	H2, H5
15 (b) (ii)	1	2.2, 2.5	H2, H5
15 (b) (iii)	2	2.3, 2.5	H2, H5
15 (c) (i)	2	14.1	H4, H5
15 (c) (ii)	2	14.1	H4, H5
15 (c) (iii)	1	14.1	H4, H5
16 (a) (i)	1	9.1	P4
16 (a) (ii)	1	1.4, 4.2	P4
16 (a) (iii)	1	11.1	H8
16 (a) (iv)	2	11.4	H8, H9
16 (b)	3	11.4, 12.3, 12.5	H3, H4, H8
16 (c) (i)	3	2.3, 2.5	P4, P5
16 (c) (ii)	4	10.6	H5